

Statistical Inference
14th of February, 2025

Introduction

Breast cancer is the prevalent issue in modern society with women being disproportionately affected by it. The earlier diagnostic rates are being associated with better survival rates which emphasizes the importance of research in this area.

This underscores the need to find new biomarkers to distinguish between benign and potentially cancerous cells, as efficient diagnosis is crucial for successful treatment. Researchers rely on morphological features to differentiate between cell types and states, particularly in cancer diagnostics

To guide the study, the following research question is considered: “Is the concavity of malignant cells greater to that of benign cells?”

The population that we are intending to make the inference about is malignant and benign cells in general.

Dataset

The Breast Cancer Wisconsin (Diagnostic) Data Set contains 569 instances of breast cancer samples, classified as either benign or malignant. It includes 30 features related to the cell characteristics, such as radius, texture, and smoothness, derived from digitized images of fine needle aspirates.

For the purpose of the study, I will be working with two columns [‘diagnosis’] which represent the variable used to make subgroups. It contains only two distinct types of records: ‘M’ and ‘B’ which stand for malignant and benign respectively. Such categorization makes it a dummy

variable, which is a nominal variable. The second column is “concavity_mean”. Concavity there is represented as a unitless continuous variable as it can take in any value within the range. ¹

Analysis

Hypothesis

To guide my further analysis to answer the research question about the concavity of malignant and benign diagnosis, the difference of means statistical and practical significance testing will be performed.

First, we have to have clearly defined null and alternative hypotheses:

- $H_0: \mu_1 - \mu_2 = 0$
 - There's no difference between the mean of the malignant cell concavity and benign cell concavity.
- $H_A: \mu_1 - \mu_2 > 0$
 - Malignant cell concavity mean is greater than the benign cell concavity mean.

Where μ_1 is the mean concavity of malignant cells, and μ_2 is the mean concavity of benign cells.

¹ #variables: As a first step for this study, the different variables are identified and classified. The variable types and roles inform the further analysis.

Notably, the test is going to be one-tailed. As mentioned in the introduction, prior research suggests that malignant tumors are generally associated with higher concavity. As a result, we are specifically testing whether the mean concavity of malignant cells is greater than the mean concavity of benign cells.

To define our significance level α , we must first consider two types of statistical errors, Type I and Type II, and assess the consequences of committing either. Since α , which is Type I error rate, serves as a threshold for when to reject or fail to reject the null hypothesis, those errors have to be put in the context of our research.

- Type I error (false positive): rejecting the null hypothesis in favor of alternative one given that the null hypothesis is true
 - Saying there's a difference, where the mean concavity of malignant cells is associated with higher values compared to benign, when in reality there's not, would lead to misdiagnosing people with breast cancer, putting emotional strain and monetary expenses on individuals.
- Type II error (false negative): failing to reject the null hypothesis in favor of alternative one given that the alternative hypothesis is true
 - Type II error would lead to misclassifying malignant tumors as benign, potentially resulting in missed diagnoses and a delay in necessary treatment. It can have serious health consequences, which could impact their prognosis and overall outcome.

Since the main goal of this study is to make an inference about whether there's a difference in mean concavity between malignant and benign cells, which can serve as a determining morphological feature in cancer diagnosis, we can proceed with a default significance level of 0.05. There's no strong prevailing reason to adjust the threshold to minimize either errors and 0.05 level will ensure that the test has a reasonable chance of detecting true effects while controlling the probability of false positives.

*Bonferroni correction is not required as there are no multiple comparisons.

Descriptive Statistics

The dataset was read into Python using the pandas package. Key statistics needed from both subgroups to perform the analysis are mean and standard deviation as we need it for all the further calculations. Also, other relevant statistics were examined such as count, skewness, kurtosis, as it will allow us to describe the distribution better and after evaluate it in terms of conditions needed for the test.

The descriptive statistics of subgroups are calculated in Appendix B using python functions that rely on algebraic expressions. The relevant summary statistics are displayed in Table 1.

Table 1: Summary statistics for the estimated cell concavity (unitless) for our two sample groups: benign and malignant diagnosis. See Appendix B for calculations.		
	Malignant cells	Benign cells

Count	$n_1 = 212$	$n_2 = 357$
Mean	$\bar{x}_1 = 0.1607747169811321$	$\bar{x}_2 = 0.04605762100840336$
Median	0.15134999999999998	0.03709
Mode	0.1204	0
Standard Deviation	$s_1 = 0.07501932785025706$	$s_2 = 0.0434421510450616$
Range	0.40282	0.4108
Skewness	0.9016281304158726	3.4732640855774015
Excess Kurtosis	1.0589955562289548	20.39576748576819

The sample size is unequal which will be important to consider in further practical analysis part when calculating pooled SD. Descriptive statistics reveal that malignant tumors have higher mean (0.161) and median (0.151) concavity than benign tumors (mean: 0.046, median: 0.037), with greater variability (std: 0.075 vs. 0.043) (“Describing data”, 2024).

The measure of excess kurtosis and skewness allows for better understanding of distribution.

The Malignant sample shows mild right skew and heavy tails, indicating some extreme values but balanced compared to the Benign sample, which has high right skew and kurtosis, suggesting clustering at lower values with extreme outliers. The skewness of both distributions emphasizes the importance of considering it when meeting the condition of normality.

To better understand the data, the following visualizations were created. See Appendix B.

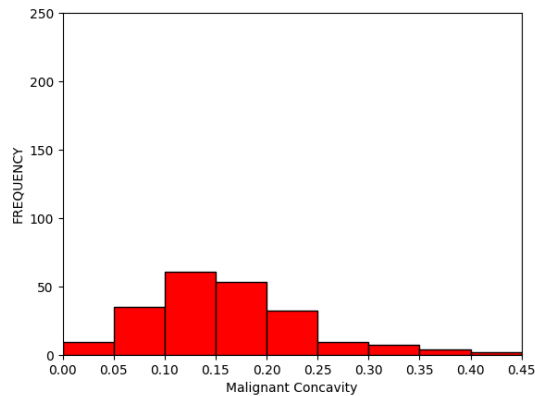


Figure 1: Histogram for malignant concavity

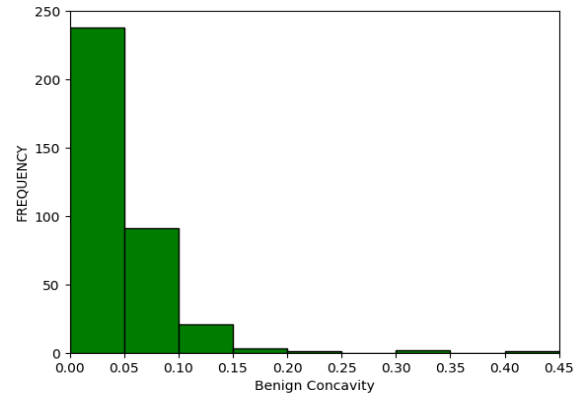


Figure 2: Histogram for benign concavity

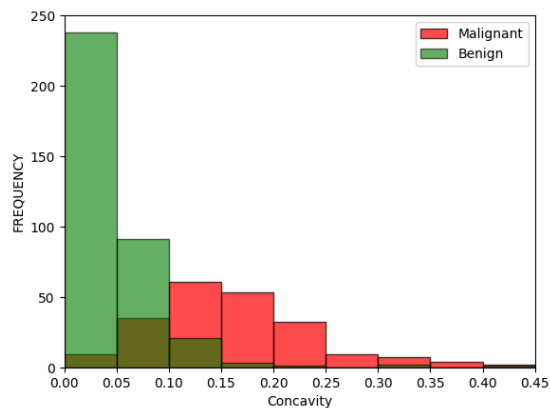


Figure 3: Histogram for both subgroups

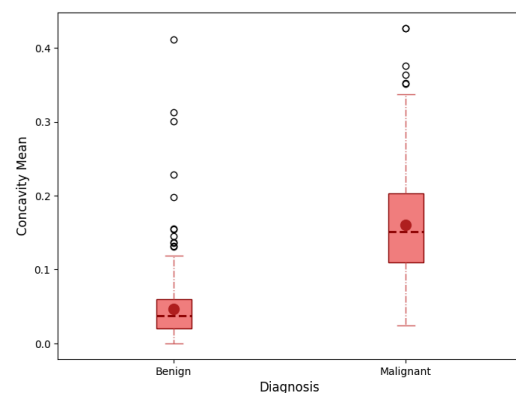


Figure 4: Box plot of concavity across subgroups²

Figure 1 and Figure 2 allow us to see the outliers and assess distribution separately, while Figure 3 visually aids in seeing the overlap between two. The histograms show that malignant tumors have a wider range of concavity values, with a concentration on the lower end but extending beyond benign values, which are clustered near zero (“Describing data”, 2024). These

² #dataviz: The histograms used allow assessing the distribution separately and see the outliers. The description outlines the strength of all visualizations.

differences are key to addressing the research question, highlighting the importance of conducting the significance testings to make the generalization about the population of cancerous and noncancerous cells.

Figure 4 is a boxplot, which illustrates the interquartile range (IQR) of the "Concavity Mean" for benign and malignant diagnoses, showing that the IQR for malignant cases is significantly higher. This highlights greater variability in concavity mean for malignant cases, which may serve as a distinguishing feature between the two diagnoses.

Conditions for t-test

We will be performing a t-test as we don't know the standard deviation of the population. Using t-test, hence t-distribution, allows us to account for uncertainty in estimating the STD of the population. In other words, t-distribution has heavier tails than the normal distribution, allowing for more extreme values, which makes it more robust to work with.

Since we are using t-test, we need to ensure that our samples meet the following criteria: randomness, independence, and normality.

- Randomness: The data collected from Fine Needle Aspiration (FNA) is based on a random sample of cells from various areas of tumors and multiple patients, ensuring randomness in the sampling process.
- Independence: Fine needle aspiration (FNA) which was used to collect the data samples cells from different areas of a tumor or from multiple patients, promoting independence (Additionally, considering that our population is malignant and benign cells in general, by

estimation, our sample size represents less than 10%, making a strong assumption for independence.

- Normality: With large sample sizes (212 for benign, 357 for malignant), CLT handles the high skewness, as visualized in sampling distribution and normal probability plot, supporting the normality assumption. See Appendix C for Python code.³

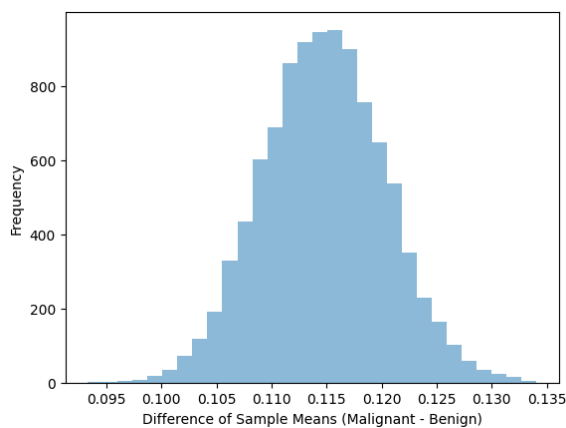


Figure 5: Sampling Distribution of Difference of Means

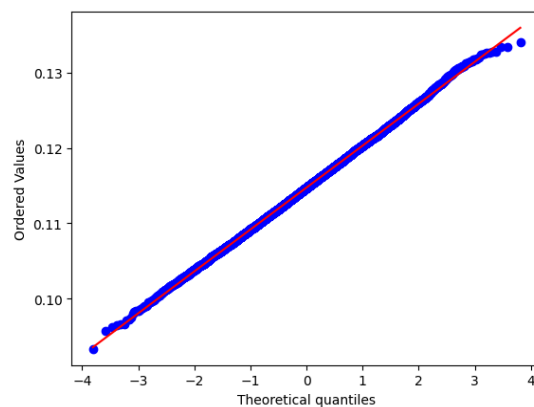


Figure 6: Normal Probability (QQ) Plot

Significance testings

To decide whether we reject or fail to reject the null hypothesis, the p-value is required to compare it with the significance level.

First, we need to compute the T-score using Welch's t-statistic formula, as the variance is not equal between subgroups. Using Python, we arrive at the following results: $T = 20.3$, with $SE = 0.0056$. The t-score indicates how many standard deviations the observed value is from the null

³ #distributions: The relevance of using t-distribution was explained as well as how it differs from normal distribution and when to use it. The sampling distribution of difference of means was explained and used to show how CLT relieves the need to ensure the normality condition. The assumption for normality was justified using the Central Limit Theorem in consideration to sample size.

hypothesis mean. Our degrees of freedom are equal to 211, which refer to the number of values in a calculation that are free to vary. See Appendix D for the calculation in Python. The T-score of 20.3 results in a one-tailed p-value of $6.56e-52^4$.

This p-value represents the probability of observing values as extreme or more extreme, given that the null hypothesis is true. Since our p-value is extremely small, it's rare to observe values as extreme as we have if the null hypothesis is true. Thus, we can reject the null hypothesis, meaning that the mean concavity of malignant and benign cells is significantly different.

To account for magnitude, in other words, to answer the question “how different?”, we need a measure of effect size. Here, we choose Cohen's d as our measure because it provides a standardized way to quantify the difference between two groups. The calculation of Cohen's d using the pooled standard deviation with unequal group sizes, as shown in Appendix D, results in an effect size of -2.003. This large negative effect size indicates a substantial practical difference between the groups. In the context of this study, this means that the difference in concavity between malignant and benign cells is not only statistically significant but also practically meaningful, with a large effect size suggesting a clear and important distinction between the two groups and can be a reliable feature for diagnosis of breast cancer. ⁵

⁴ #probability: Information is being appropriately interpreted treating p-value as the probability of obtaining events as extreme as observed or more. When working with confidence intervals, frequentist interpretation of probability to arrive at conclusion.

⁵ #significance: A measure of significance was interpreted to explain the observed statistical difference. Statistical significance was distinguished from the practical significance of the results, the implications of both in the context of study were considered. Step-by-step process is shown to demonstrate what lies in the foundation of Cohen's d - the measure of practical significance and p-value - the measure of statistical significance.

Confidence intervals

To continue with our analysis, we can construct confidence intervals. These intervals will reflect the precision of our estimates for each group's mean. We set our confidence level to 95% because it provides a high level of certainty about the estimates.

To compute the confidence intervals, we use the general formula for a 95% confidence interval and the standard error with formula $= \text{std}/n^{0.5}$. The full calculation is found in Appendix C. The resulting 95% confidence intervals are: Malignant cells: [0.15, 0.17] Benign cells: [0.04, 0.05]

We can be 95% confident that the true mean concavity of malignant cells falls between 0.15 and 0.17, while the true mean for benign cells falls between 0.04 and 0.05. This means that, in line with frequentist interpretation of probability, we are 95% confident the population means lie within these intervals. The fact that the intervals do not overlap suggests a significant difference in concavity between malignant and benign cells, which is expected after seeing the results of statistical and practical significance testings.⁶

Results and Conclusions

To conclude, malignant cells have a strong tendency to exhibit greater concavity compared to benign cells both from statistical and practical standpoint (p-value = $6.56e-52$, Cohen's $d = 2.003$). This finding supports the use of cell concavity as a reliable biomarker for breast cancer

⁶ #confidenceintervals: Upper and lower values that are likely to contain the population parameter, the difference in concavity in two subgroups, were calculated. The choice of the confidence interval of 95% was justified. The intervals for two subgroups are interpreted correctly and are used to inform the conclusion about alternative hypotheses as lack of overlap was observed. The results matched with what was expected after significance testing emphasizing the tight connection between two.

diagnosis, aligning with the need for efficient early detection methods to improve the efficiency of treatment and decrease mortality. The substantial effect size indicates a clear distinction between malignant and benign cells, potentially improving diagnostic accuracy. Future research should focus on integrating this concavity measure into automated diagnostic tools and exploring its effectiveness in combination with other cellular features for more comprehensive breast cancer screening.⁷

Important to consider, that while study shows strong evidence that supports the introduced alternative hypothesis, there's always room for Type I and Type II errors.⁸⁹

⁷ #induction: The conclusion serves as a strong example of induction as it makes the generalization about the population, which is in this context, the difference in concavity between malignant and benign cells, from limited data - sample. The conclusion emphasizes the strength and reliability of the induction by mentioning the statistical tools used. It goes further by outlining the limitations of generalization that are always present in induction such as the chance of still facing Type I or Type II error. Here, we assess the observations and make the conclusion that is likely to be true.

⁸ #organization: The study has an easy-to-follow format where each part is being built on top of each other. The titles are used for easier navigation.

References

- “Describing data” [November 10, 2024]
- Breast Cancer Wisconsin (Diagnostic) Data Set.
https://course-resources.minerva.edu/uploaded_files/mu/00353110-6347/breast-cancer-wisconsin.csv

Reflections:

1.

2.

Appendix A - Import of data

All used libraries were imported in this cell.

...

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import random
from scipy import stats
from pprint import pprint #print complex data structures in easy-to-read format
```

```
[ ] df = pd.read_csv("https://course-resources.minerva.edu/uploaded_files/mu/00353110-6347/breast-cancer-wisconsin.csv")
df.head()
df = df.drop(columns=['id'], inplace=False)
df.head()
```



	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	compactness_mean	concavity_mean	concave points_mean	symmetry_mean	...
0	M	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	0.14710	0.2419	...
1	M	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	0.07017	0.1812	...
2	M	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	0.12790	0.2069	...
3	M	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	0.10520	0.2597	...
4	M	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	0.10430	0.1809	...

5 rows x 32 columns

```
[ ] if df['concavity_mean'].isnull().values.any():
    print("The 'concavity_mean' column has NaN values.")
else:
    print("The 'concavity_mean' column does not have NaN values.")
```



The 'concavity_mean' column does not have NaN values.

Appendix B - Descriptive Statistics and visualizations

The code was adapted from "Describing Data" assignment.

```
'''

def my_mean(data): #passes the list or array of numerical values
    sumtotal = 0
    n = len(data)

    for element in data: #iterates over each element in the dataset
        sumtotal += element

    return sumtotal/n #mean formula

def my_median(data):

    #sort the values in the dataset from the smallest to the greatest
    sorted_data = sorted(data)
    n = len(sorted_data)

    if n % 2 == 1: #odd number of elements
        return sorted_data[n//2] #index of the middle element

    else: #even number of elements
        middle_1, middle_2 = n//2 - 1, n//2 # "n//2 - 1" index of the first middle element, " n//2" index of the second middle element.
        return (sorted_data[middle_1] + sorted_data[middle_2]) / 2 #average of two middle elements

def my_count(data, item):

    occurrences = 0
    for element in data: #iteration over each element in the dataset
        if element == item:
            occurrences += 1
    return occurrences
```

```
def my_mode(data):
    max_count = 0

    max_items = [] #the empty list where mode (or modes) will be stored

    for element in data:
        current_count = my_count(data, element) #counts number of the same element in the dataset.

        if current_count > max_count:
            max_count = current_count #the max_count will automatically be updated with new numerical value stored in it
            max_items = [element] #the variable assigns the elements that appeared with the highest frequency so far in the dataset
        elif current_count == max_count and element not in max_items:
            max_items.append(element)

    if max_count == 1:
        return None
    return max_items if len(max_items) > 1 else max_items[0]

def my_sample_SD(data):
    sum_of_squares = 0
    n = len(data)
    mean = my_mean(data) #mean is part of standart deviation formula

    for element in data: #iteration over each element in the dataset
        sum_of_squares += (element - mean)**2 #the key expression in sample standart deviation formula to the sum)

    return (sum_of_squares/(n-1))*0.5 #final operations for std formula

def my_range(data):

    sorted_data = sorted(data)
    return sorted_data[-1] - sorted_data[0] #indexing that extracts last and first value in the dataset
```

```

def my_skewness(data):

    n = len(data)
    if n < 3:
        raise ValueError("Skewness requires at least 3 data points.")

    mean = my_mean(data)
    std = my_sample_SD(data)

    # Calculate skewness using algebraic formula
    skewness = (n / ((n - 1) * (n - 2))) * np.sum(((data - mean) / std) ** 3)

    return skewness


def my_kurtosis(data):

    n = len(data)
    mean = my_mean(data) #compute the mean
    std = my_sample_SD(data) #compute SD

    # Calculate the fourth central moment
    fourth_moment = np.sum((data - mean)**4) / n

    # Calculate the kurtosis
    kurtosis = fourth_moment / (std**4)

    # Excess kurtosis (normal distribution has excess kurtosis = 0)
    excess_kurtosis = kurtosis - 3

    return excess_kurtosis

```

The code was adapted from "Describing Data" assignment.

```
'''

def descriptive_statistics(df):
    malignant_concavity = df[df['diagnosis'] == 'M']['concavity_mean']
    benign_concavity = df[df['diagnosis'] == 'B']['concavity_mean']

    statistics = {
        'Malignant': {
            'Mean': my_mean(malignant_concavity),
            'Median': my_median(malignant_concavity),
            'Mode': my_mode(malignant_concavity),
            'Standard Deviation': my_sample_SD(malignant_concavity),
            'Range': my_range(malignant_concavity),
            'Skewness': my_skewness(malignant_concavity),
            'Excess_Kurtosis': my_kurtosis(malignant_concavity)
        },
        'Benign': {
            'Mean': my_mean(benign_concavity),
            'Median': my_median(benign_concavity),
            'Mode': my_mode(benign_concavity),
            'Standard Deviation': my_sample_SD(benign_concavity),
            'Range': my_range(benign_concavity),
            'Skewness': my_skewness(benign_concavity),
            'Excess_Kurtosis': my_kurtosis(benign_concavity)
        }
    }

    return statistics

statistics = descriptive_statistics(df)

pprint(statistics, sort_dicts=False)
```

```
{'Malignant': {'Mean': 0.1607747169811321,
               'Median': 0.15134999999999998,
               'Mode': 0.1204,
               'Standard Deviation': 0.07501932785025706,
               'Range': 0.40282,
               'Skewness': 0.9016281304158726,
               'Excess_Kurtosis': 1.0589955562289548},
 'Benign': {'Mean': 0.04605762100840336,
            'Median': 0.03709,
            'Mode': 0.0,
            'Standard Deviation': 0.0434421510450616,
            'Range': 0.4108,
            'Skewness': 3.4732640855774015,
            'Excess_Kurtosis': 20.39576748576819}}
```

$$Skewness = \frac{n}{(n-1)(n-2)} \sum_{i=1}^n \left(\frac{x_i - mean}{Stdev} \right)^3$$

Excess Kurtosis: $\frac{\mu^4}{\sigma^4} - 3$

*Excess kurtosis allows us to compare the distribution's kurtosis to that of a normal distribution;

Since the kurtosis of a normal distribution is 3, we subtract 3 to make it "excess" kurtosis).

Hence, if the distribution is normal, the excess kurtosis will be 0.

Forth central moment: $\mu_4 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^4$

The **central moment** of a dataset measures how the data points are spread out or distributed around the mean. The **n-th central moment** gives a sense of the shape of the distribution in terms of how much the data points deviate from the mean raised to the power of n.

- **First central moment: Mean**
- **Second central moment: Variance**
- **Third central moment: Skewness** (Measures the asymmetry of the distribution. A positive third central moment means the data is skewed to the right (long tail on the right), and a negative one means the data is skewed to the left)
- **Fourth central moment: Kurtosis** (Measures the "tailedness" of the distribution — how extreme the outliers are. Higher values indicate heavier tails)

```
[ ] '''
    The code was adapted from "Decribing Data" assignment.
    ...

    $two separate histograms

    $without scaling (Malignant)
    plt.hist(malignant_concavity, bins=[0.0, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5,0.55, 0.6, 0.65, 0.7, 0.75, 0.8, 0.85, 0.9, 0.95, 1.0], edgecolor="black", color = "red")
    plt.xlabel("Malignant Concavity")
    plt.xlim(0, 0.45) #given the range of 0.40282
    plt.ylabel("FREQUENCY")

    plt.show()

    $with scaling (Malignant)
    plt.hist(malignant_concavity, bins=[0.0, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5,0.55, 0.6, 0.65, 0.7, 0.75, 0.8, 0.85, 0.9, 0.95, 1.0], edgecolor="black", color = "red")
    plt.xlabel("Malignant Concavity")
    plt.ylim(0, 250)
    plt.xlim(0, 0.45) #given the range of 0.40282
    plt.ylabel("FREQUENCY")

    plt.show()

    plt.hist(benign_concavity, bins=[0.0, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5,0.55, 0.6, 0.65, 0.7, 0.75, 0.8, 0.85, 0.9, 0.95, 1.0], edgecolor="black", color = "green")
    plt.xlabel("Benign Concavity")
    plt.ylim(0, 250)
    plt.xlim(0, 0.45) #given the range of 0.4108
    plt.ylabel("FREQUENCY")
    plt.show()
```

```
#both histograms displayed on the single set of axes
plt.hist(malignant_concavity, bins=[0.0, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5, 0.55, 0.6, 0.65, 0.7, 0.75, 0.8, 0.85, 0.9, 0.95, 1.0], edgecolor="black", color = "red", alpha = 0.7, label = "Malignant")
plt.hist(benign_concavity, bins=[0.0, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5, 0.55, 0.6, 0.65, 0.7, 0.75, 0.8, 0.85, 0.9, 0.95, 1.0], edgecolor="black", color = "green", alpha = 0.6, label = "Benign")
plt.xlabel("Concavity")
plt.ylim(0, 250)
plt.xlim(0, 0.45)
plt.ylabel("FREQUENCY")

plt.legend()
plt.show()
```

```
plt.figure(figsize=(8, 6))
data = [benign_concavity, malignant_concavity]
labels = ['Benign', 'Malignant']

plt.boxplot(data, labels=labels, patch_artist=True, showmeans=True,
            meanprops={"marker": "o", "markerfacecolor": "firebrick", "markeredgecolor": "firebrick", "markersize": 10},
            medianprops={"linestyle": "--", "color": "darkred", "linewidth": 2},
            boxprops={"facecolor": "lightcoral", "color": "darkred"},
            whiskerprops={"color": "indianred", "linestyle": "-."},
            capprops={"color": "indianred"})

plt.xlabel("Diagnosis", fontsize=12)
plt.ylabel("Concavity Mean", fontsize=12)

plt.show()
```

Appendix C: T-test conditions

```
sample_size = len(malignant_concavity) + len(benign_concavity)

sample_means_malignant = []
sample_means_benign = []

for _ in range(10000): # Adjust the number of samples as needed
    # Resample with replacement to create the sampling distribution
    # Convert Pandas Series to lists using .tolist()
    malignant_sample = random.choices(malignant_concavity.tolist(), k=len(malignant_concavity))
    benign_sample = random.choices(benign_concavity.tolist(), k=len(benign_concavity))

    sample_means_malignant.append(np.mean(malignant_sample))
    sample_means_benign.append(np.mean(benign_sample))
```

```
differences = np.array(sample_means_malignant) - np.array(sample_means_benign)

# Visualize the sampling distribution of the difference of means
plt.hist(differences, bins=30, alpha=0.5, )
plt.xlabel('Difference of Sample Means (Malignant - Benign)')
plt.ylabel('Frequency')
plt.show()

# Normal probability plot for the difference of means
res = stats.probplot(differences, plot=plt)
plt.title('')
plt.show()
```

Appendix D: Significance testings



```
n1_m = len(malignant_concavity)
n2_b = len(benign_concavity)

x1_m = np.mean(malignant_concavity)
x2_b = np.mean(benign_concavity)

s1_m = np.std(malignant_concavity, ddof=1) # sample SD formula (aka Bessel's correction): use n-1 in denominator
s2_b = np.std(benign_concavity, ddof=1)

standart_error = np.sqrt(s1_m**2/n1_m + s2_b**2/n2_b)
t_score = np.abs((x2_b - x1_m)/standart_error)
df = min(n1_m, n2_b) - 1 # conservative estimate from OpenIntro

print("SE:", standart_error)
print("t-score:", t_score)
print("df:", df)
```



```
SE: 0.005642076558618796
t-score: 20.332424556963463
df: 211
```

```
[ ]
```

```
'''
```

```
The code was adapted from FA50 Session 23 - (13.2) Difference of Means Tests and Effect Size
```

```
'''
```

The code was adapted from FA50 Session 23 - (13.2) Difference of Means Tests and Effect Size

```
'''  
  
def difference_of_means_test(data1,data2,tails):  
    n1 = len(data1)  
    n2 = len(data2)  
  
    x1 = np.mean(data1)  
    x2 = np.mean(data2)  
  
    s1 = np.std(data1,ddof=1) # sample SD formula (aka Bessel's correction): use n-1 in denominator  
    s2 = np.std(data2,ddof=1)  
  
    standard_error = np.sqrt(s1**2/n1 + s2**2/n2)  
    t_score = np.abs((x2 - x1))/standard_error  
    df = min(n1,n2) - 1 # conservative estimate from OpenIntro  
    p_value = tails*stats.t.cdf(-t_score,df) # for a 1 or 2-tailed test  
  
    sd_pooled = np.sqrt((s1**2*(n1-1) + s2**2*(n2-1))/(n1+n2-2)) #unequal group sizes  
    cohens_d = (x2 - x1)/sd_pooled  
  
    print('p =',p_value)  
    print('d =',cohens_d)  
  
print("Difference-of-means function loaded and ready to use.")
```

☞ Difference-of-means function loaded and ready to use.

```
] difference_of_means_test(malignant_concavity, benign_concavity, 1)
```

☞ p = 6.5566013241717336e-52
d = -2.0032750549138316

$$t' = \frac{\mu_1 - \mu_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}.$$

$$SD^*_{pooled} = \sqrt{\frac{(n_1 - 1)SD_1^2 + (n_2 - 1)SD_2^2}{n_1 + n_2 - 2}}$$

Where:

- $\bar{x}$ = the sample mean
- n_1 = sample size for group 1
- n_2 = sample size for group 2

```
#malignant concavity confidence interval

SE_m = s1_m/n1_m**0.5

t = stats.t.ppf(0.975,n1_m-1) # critical t value

print("Threshold t value for 95% confidence =",round(t,2))
print("Confidence interval of malignant concavity:", [round(x1_m-t*SE_m,2),round(x1_m+t*SE_m,2)])
```

```
Threshold t value for 95% confidence = 1.97
Confidence interval of malignant concavity: [0.15, 0.17]
```

```
#benign concavity confidence interval
```

```
SE_m = s2_b/n2_b**0.5

t = stats.t.ppf(0.975,n2_b-1) # critical t value

print("Threshold t value for 95% confidence =",round(t,2))
print("Confidence interval of malignant concavity:", [round(x2_b-t*SE_m,2),round(x2_b+t*SE_m,2)])
```

```
Threshold t value for 95% confidence = 1.97
Confidence interval of malignant concavity: [0.04, 0.05]
```

Appendix E - [AI statement]

Grammarly and Wordtune were used for sentence-level editing. ChatGPT was used for debugging the code to visualize sampling distribution .